

MATT WOERMAN

CONTACT INFORMATION

Agricultural and Resource Economics
Colorado State University
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ACADEMIC APPOINTMENTS

Assistant Professor. Agricultural and Resource Economics. Colorado State University. 2023–present
Assistant Professor. Resource Economics. University of Massachusetts Amherst. 2019–2023

EDUCATION

Ph.D. Agricultural and Resource Economics. University of California, Berkeley. 2019
M.S. Agricultural and Resource Economics. University of California, Berkeley. 2015
M.S. Agricultural, Environmental, and Development Economics. The Ohio State University. 2009
B.S. Mechanical Engineering. Kansas State University. 2007

RESEARCH INTERESTS

Environmental and natural resource economics, energy economics, and industrial organization

PEER-REVIEWED PUBLICATIONS

Residential demand for energy in light of changing solar prices (with Rafael Bakhtavoryan and Vardges Hovhannisyan). *Energy Economics*, 144: 108351. 2025.

Intermittency and uncertainty in wind and solar energy: Impacts on the French electricity market (with Maurice K. Ekoue and Cédric Clastres). *Energy Economics*, 142: 108176. 2025.

Intermittency or uncertainty? Impacts of renewable energy in electricity markets (with Paige Weber). *Journal of the Association of Environmental and Resource Economists*, 11 (6): 1351–1385. 2024. Winner of the 2022 USAEE Young Professional Best Paper Award.

Linking carbon markets with different initial conditions. *Journal of Environmental Economics and Management*, 119: 102820. 2023.

Panel data and experimental design (with Fiona Burlig and Louis Preonas). *Journal of Development Economics*, 144: 102458. 2020.

Flexibility and stringency in greenhouse gas regulations (with Dallas Burtraw and Alan Krupnick). *Environmental and Resource Economics*, 63(2): 225–248. 2016.

Incentives, margins, and cost effectiveness in comprehensive climate policy for the power sector (with Anthony Paul and Karen Palmer). *Climate Change Economics*, 6(4): 1–27. 2015.

Analysis of the Bingaman Clean Energy Standard proposal (with Anthony Paul and Karen Palmer). *Resource and Energy Economics*, 36(1): 113–129. 2014.

Economic ideas for a complex climate policy regime (with Dallas Burtraw). *Energy Economics*, 40 (Supplement 1): S24–S31. 2013.

Reliability in the U.S. electricity industry under new environmental regulations (with Dallas Burtraw, Karen Palmer, Anthony Paul, and Blair Beasley). *Energy Policy*, 62: 1078–1091. 2013.

Modeling a clean energy standard for electricity: Policy design implications for emissions, supply, prices and regions (with Anthony Paul and Karen Palmer). *Energy Economics*, 36: 108–124. 2013.

Cost-effectiveness and economic incidence of a clean energy standard (with Bryan K. Mignone, Thomas Alfstad, Aaron Bergman, Kenneth Dubin, Richard Duke, Paul Friley, Andrew Martinez, Matthew Mowers, Karen Palmer, Anthony Paul, Sharon Showalter, Daniel Steinberg, and Frances Wood). *Economics of Energy and Environmental Policy*, 1(3): 59–86. 2012.

Secular trends, environmental regulations, and electricity markets (with Dallas Burtraw, Karen Palmer, and Anthony Paul). *The Electricity Journal*, 25(6): 35–47. 2012.

Retail electricity price savings from compliance flexibility in GHG standards for stationary sources (with Dallas Burtraw and Anthony Paul). *Energy Policy*, 42: 67–77. 2012.

Strategies for mitigating climate change through energy efficiency: The RFF Haiku electricity market model (with Anthony Paul and Karen Palmer). *The Energy Journal*, 32 (Special Issue): 193–212. 2011.

Federal policies for renewable electricity: Impacts and interactions (with Karen Palmer, Anthony Paul, and Daniel C. Steinberg). *Energy Policy*, 39(7): 3975–3991. 2011.

WORKING PAPERS

Groundwater and crop choice in the short and long run (with Fiona Burlig and Louis Preonas). NBER Working Paper 28706. Revisions requested, *Review of Economic Studies*.

Market size and market power: Evidence from the Texas electricity market. Revisions requested, *RAND Journal of Economics*.

Mass reproducibility and replicability: A new hope (with Abel Brodeur, Derek Mikola, Nikolai Cook et al.). I4R Discussion Paper No. 107. Revisions requested, *Nature*.

Unintended benefits of large irrigation projects in India (with Prashikdivya Gajbhiye and Nilesh Shinde). Revisions requested, *Journal of Development Economics*.

SELECTED WORK IN PROGRESS

Groundwater in a changing climate (with Fiona Burlig, R. Aaron Hrozencik, and Louis Preonas)

Gradual linking of carbon markets across space and time (with Jan Abrell, Dallas Burtraw, Michael Pahle, and Simon Queminn)

Pricing the pump: Demand response to two-part electricity tariffs (with Fiona Burlig and Louis Preonas)

Internet suppliers or DSO: Which technologies and networks to supply energy services? (with Cédric Clastres and Rania Mabrouk)

OTHER PUBLICATIONS AND MEDIA

Yellow vests, pessimistic beliefs, and carbon tax aversion (2022): A comment (with Nicholas Rivers and Kareman Yassin). I4R Discussion Paper No. 58. 2023.

Water shortages threaten the economy and our food supply (with Fiona Burlig and Louis Preonas). *Barron's*, October 4, 2021.

Groundwater use in California agriculture (with Fiona Burlig and Louis Preonas). *Forecast Direct podcast*, UCLA Anderson School of Management, July 17, 2021.

Mercury and Air Toxics Standards analysis deconstructed: Changing assumptions, changing results (with Blair Beasley, Anthony Paul, Dallas Burtraw, and Karen Palmer). RFF Discussion Paper 13-10. 2013.

Linking by degrees: Incremental alignment of cap-and-trade markets (with Dallas Burtraw, Karen Palmer, Clayton Munnings, and Paige Weber). RFF Discussion Paper 13-04. 2013.

Clean air regulations and the electricity sector (with Karen Palmer, Dallas Burtraw, Anthony Paul, and Blair Beasley). *Resources Magazine*, Fall 2012.

US status on climate change mitigation (with Dallas Burtraw). RFF Discussion Paper 12-48 and Mistra Indigo Policy Paper 1. 2012.

The effect of natural gas supply on retail electricity prices (with Karen Palmer, Dallas Burtraw, and Blair Beasley). RFF Issue Brief 12-05. 2012.

The variability of potential revenue from a tax on carbon (with Karen Palmer and Anthony Paul). RFF Issue Brief 12-03. 2012.

Supply curves for conserved electricity (with Anthony Paul and Karen Palmer). RFF Discussion Paper 11-11. 2011.

GRANTS

USDA NIFA Agricultural and Food Research Initiative grant. “The economics of climate, groundwater, and agriculture: Empirical evidence from US cropland” (with Fiona Burlig, R. Aaron Hrozencik, and Louis Preonas). \$650,000. 2024–2028

Potsdam Institute for Climate Impact Research and Resources for the Future research funding. “Gradual linking of carbon markets across space and time.” \$18,000. 2025

USDA ERS cooperative agreement. “Climate adaptation in irrigated agriculture: Impact of short-run weather shocks and groundwater scarcity on water and land use decisions” (with Fiona Burlig and Louis Preonas). \$150,000. 2020–2023

Alfred P. Sloan Foundation E2e project grant. “Energy efficiency in agriculture: Energy use, water demand, and welfare” (with Fiona Burlig and Louis Preonas). \$222,289. 2018–2021

J-PAL North America project development grant. “Agricultural irrigation efficiency: Energy, water, and profits” (with Fiona Burlig and Louis Preonas). \$4,752. 2018–2019

Giannini Foundation mini-grant. “Estimating the effect of agricultural pumping efficiency upgrades on farmers’ electricity consumption, water consumption, crop production, and profits” (with Fiona Burlig and Louis Preonas). \$26,882. 2017–2018

Berkeley Institute for Transparency in the Social Sciences SSMART grant. “Power calculations and clustering: New techniques for transparent social science research” (with Fiona Burlig and Louis Preonas). \$29,355. 2016

PRESENTATIONS

2026 (including planned): Colorado School of Mines (Economics & Business); WEAI Annual Conference

2025: University of New Mexico (Economics); AERE Summer Conference; Big Sky Workshop on the Economics of Agriculture and the Environment; Boulder Energy, Resilience, and Sustainability Summer Workshop; Front Range Energy and Environmental Economics Camp

2024: University of California, Berkeley (Energy Institute at Haas webinar); University of Wyoming (Agricultural and Applied Economics); AERE Summer Conference; ASSA Annual Meeting (discussant); Front Range Energy and Environmental Economics Camp

2023: Colorado State University (Agricultural and Resource Economics); ASSA Annual Meeting; Colorado Environmental and Resource Economics Workshop

2022: University of Rhode Island (Environmental and Natural Resource Economics); Yale (Environmental and Energy Economics Seminar); U.S. Department of Agriculture, Economic Research Service; AAEE Annual Meeting; Energy Institute at Haas Energy Camp; EPIC Global Energy Challenge Conference; Northeast Workshop on Energy Policy and Environmental Economics

2021: AERE Summer Conference; ASSA Annual Meeting; EAERE Annual Conference; NAREA Annual Meeting; Online Summer Workshop in Environment, Energy, and Transportation; RFF-PIK Climate Policy Workshop; Seminar in Water Economics Online

2020: AERE Summer Conference; ASSA Annual Meeting

2019: Case Western Reserve University (Economics); Rutgers University (Economics); Santa Clara University (Economics); University of Alberta (Economics); University of Connecticut (Agricultural and Resource Economics); University of Maryland, College Park (Agricultural and Resource Economics); University of Massachusetts Amherst (Resource Economics); University of Utah (Finance); U.S. Department of Justice, Antitrust Division; AERE Summer Conference; Colorado Environmental and Resource Economics Workshop; Energy Institute at Haas Energy Camp; Heartland Environmental and Resource Economics Workshop

2018: University of California, Berkeley (Agricultural and Resource Economics); University of Nevada, Reno (Economics)

2017: AERE Summer Conference; Energy Institute at Haas Energy Camp

2016: University of California, Berkeley (Environmental and Resource Economics Seminar); AERE Summer Conference; Heartland Environmental and Resource Economics Workshop; Interdisciplinary PhD Workshop in Sustainable Development

2015: University of California, Berkeley (Environmental and Resource Economics Seminar); Berkeley Initiative for Transparency in the Social Sciences Annual Meeting; Energy Institute at Haas Energy Camp; Occasional Workshop in Environmental and Resource Economics (discussant)

Previous: Resources for the Future (2012); Trans-Atlantic Infraday Conference (2010, 2011, 2012)

RESEARCH EXPERIENCE

Graduate student researcher. Energy Institute at Haas. University of California, Berkeley. 2014–2019

Research associate. Resources for the Future. 2012–2013

Senior research assistant. Resources for the Future. 2011–2012

Research assistant. Resources for the Future. 2009–2011

TEACHING EXPERIENCE

Colorado State University

AREC 230: Agricultural Data Management and Analysis (undergraduate introductory course)

Semesters: Spring 2024

AREC 540: Environmental and Natural Resource Economics (masters course)

Semesters: Spring 2024, Spring 2025

AREC 542: Applied Advanced Water Resource Economics (masters course)

Semesters: Spring 2025

University of Massachusetts Amherst

ResEcon 262: Environmental Economics (undergraduate general education course)

Semesters: Spring 2023

Most recent instructor rating: 4.5/5

ResEcon 428: Managerial Economics (undergraduate capstone course)

Semesters: Spring 2020, Spring 2021, Fall 2022

Most recent instructor rating: 4.8/5

ResEcon 703: Topics in Advanced Econometrics (graduate course)

Semesters: Fall 2019, Fall 2020, Fall 2021, Fall 2022

Most recent instructor rating: 5.0/5

ADVISING

Maurice Ekoue, Université Grenoble Alpes. PhD committee member. Expected 2028
Jaeyong Jung, Colorado State University. PhD committee member. Expected 2028
Shuhang Lou, Colorado State University. PhD committee member. Expected 2028
Anup Sharma, Colorado State University. PhD committee member. Expected 2028
Dan Ai, University of Massachusetts Amherst. PhD committee member. Expected 2027
Bhishma Dahal, Colorado State University. PhD committee chair. Expected 2027
Joseph Huyett, Colorado State University. PhD committee member. Expected 2027
Xiaolin Zhou, University of Massachusetts Amherst. PhD committee member. Expected 2027
Prashikdivya Gajbhiye, University of Massachusetts Amherst. PhD committee co-chair. Expected 2026
Dhijoj Koirala, University of Massachusetts Amherst. PhD committee co-chair. Expected 2026
Casey Buchholz, University of Massachusetts Amherst. PhD committee member. 2025
Osung Kwon, University of Massachusetts Amherst. PhD committee member. 2025
Joshua Budlender, University of Massachusetts Amherst. PhD committee member. 2024
Ming Ge, University of Massachusetts Amherst. PhD committee member. 2023
Mateo Hoyos, University of Massachusetts Amherst. PhD committee member. 2023
Camilo Gallego, University of Massachusetts Amherst. PhD committee member. 2022
Eli Miller, University of Massachusetts Amherst. MS field essay advisor. 2021

EDITORIAL SERVICE

Editorial Council. *Journal of Agricultural and Resource Economics*. 2025–present

REFeree SERVICE

American Journal of Agricultural Economics, Climate Policy, Economic Development and Cultural Change, Energy Economics, Energy Policy, European Review of Agricultural Economics, Journal of Benefit-Cost Analysis, Journal of Environmental Economics and Management, Journal of Political Economy Microeconomics, Journal of Public Economics, Journal of the Association of Environmental and Resource Economists, Nature Climate Change, Resource and Energy Economics, SN Applied Sciences, Water Resources Research

Excellence in Refereeing Award. *Journal of the Association of Environmental and Resource Economists*. 2026.

PROFESSIONAL SERVICE

Review panel. USDA NIFA AFRI Economics, Markets and Trade. 2026 (planned)
Abstract review. AAEA Annual Meeting. 2023–2025
Abstract review. AERE Summer Conference. 2023–2025
Abstract review. EEM International Conference. 2025
Abstract review. IAEE International Conference. 2025
Abstract review. WAEA Annual Meeting. 2024–2025
Abstract review. NAREA Annual Meeting. 2022–2023
Grant review. Social Sciences and Humanities Research Council of Canada. 2023
Session organizer. NAREA Annual Meeting. 2021
Session chair. EAERE Annual Conference. 2021
Session chair. IAEE International Conference. 2021
Session chair. NAREA Annual Meeting. 2021
Session chair. AERE Summer Conference. 2020

DEPARTMENT SERVICE

Colorado State University

Econometrics qualifying exam committee member. 2024–present

ENRE (Environmental and Natural Resource Economics) Club advisor. 2023–present

Undergraduate committee member. 2023–present

University of Massachusetts Amherst

Hiring committee member. 2021, 2023

Diversity, equity, and inclusion committee member. 2020–2023

Seminar committee chair. 2022–2023

Environment and natural resource economics reading group organizer. 2021–2023

Econometrics comprehensive exam committee member. 2019–2023

Seminar committee member. 2019–2022

Graduate studies committee member. 2020–2021